

HU Collection

TMUA Mock Set H Paper 1

Worked Solutions

Kevin Hu 24 September 2026

Answer key

Q	1	2	3	4	5	6	7	8	9	10
Ans	D	B	E	A	C	B	F	D	A	C
Q	11	12	13	14	15	16	17	18	19	20
Ans	E	B	A	C	D	F	E	B	C	D

Each solution gives the intended route and names the trap the question was built around. Questions are ordered by increasing difficulty.

Question 1. Answer D.

The function is piecewise linear, so track its gradient. The kinks are at $1, \frac{3}{2}$ and 2 , where the gradient jumps by $2, 4$ and 6 respectively (twice the coefficient of x). Far to the left the gradient is $-(1 + 2 + 3) = -6$; it becomes -4 after $x = 1$, 0 after $x = \frac{3}{2}$, and 6 after $x = 2$. So the function is constant on $[\frac{3}{2}, 2]$, and that constant is the minimum: at $x = 2$ it is $1 + 1 + 0 = 2$. The minimum sits where the “weight” to the left first reaches half the total, the weighted median.

Question 2. Answer B.

Take reciprocals: $\frac{1}{a_{n+1}} = \frac{1+na_n}{a_n} = \frac{1}{a_n} + n$. So the reciprocals increase by $1, 2, 3, \dots$ in turn, and

$$\frac{1}{a_{100}} = \frac{1}{a_1} + (1 + 2 + \dots + 99) = 1 + 4950 = 4951.$$

Computing the first few terms directly gives $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{7}, \dots$, which is how the reciprocal pattern suggests itself.

Question 3. Answer E.

Complete the square: multiplying by 4 gives $(2x + y)^2 + 3y^2 = 28$, so $3y^2 \leq 28$ and $|y| \leq 3$. $y = \pm 3$: $(2x \pm 3)^2 = 1$, giving $x = -1, -2$ for $y = 3$ and $x = 1, 2$ for $y = -3$. $y = \pm 2$: $(2x \pm 2)^2 = 16$, giving $x = 1, -3$ for $y = 2$ and $x = -1, 3$ for $y = -2$. $y = \pm 1$: $(2x \pm 1)^2 = 25$, giving $x = 2, -3$ for $y = 1$ and $x = -2, 3$ for $y = -1$. $y = 0$: $4x^2 = 28$, no integer solutions. That is 12 pairs, matching the six-fold symmetry of the equation.

Question 4. Answer A.

Divide by 4^x : $(\frac{3}{2})^{2x} - (\frac{3}{2})^x - 1 = 0$. With $t = (\frac{3}{2})^x > 0$, $t^2 - t - 1 = 0$, so $t = \frac{1+\sqrt{5}}{2}$. Then $(\frac{2}{3})^x = \frac{1}{t}$, and $t - \frac{1}{t} = 1$ comes straight from $t^2 - t = 1$, no need to find x or even t . The quantity is 1 .

Question 5. Answer C.

By the cosine rule, the smallest angle A (opposite 4) has $\cos A = \frac{25+36-16}{60} = \frac{3}{4}$, and the largest angle C (opposite 6) has $\cos C = \frac{16+25-36}{40} = \frac{1}{8}$. Now $2\cos^2 A - 1 = \frac{9}{8} - 1 = \frac{1}{8} = \cos C$. Using

$\cos 2A = 2\cos^2 A - 1$, and since $2A$ and C both lie in $(0^\circ, 180^\circ)$, where cosine is one-to-one, we get $C = 2A$. The ratio is exactly 2, which no amount of approximating the angles would reveal with certainty.

Question 6. Answer B.

Each tangent at P is perpendicular to the radius there, so perpendicular tangents mean perpendicular radii: triangle O_1PO_2 has a right angle at P , and $O_1O_2 = \sqrt{9+25} = \sqrt{34}$. The chord PQ is bisected at right angles by O_1O_2 , and half of it is the altitude from P in that right triangle: $\frac{3 \cdot 5}{\sqrt{34}}$. So $PQ = \frac{30}{\sqrt{34}} = \frac{15\sqrt{34}}{17}$.

Question 7. Answer F.

Substituting $y = x^2$ into $x^2 + y^2 + Dx + Ey + F = 0$ gives the quartic $x^4 + (1+E)x^2 + Dx + F = 0$, which has no x^3 term. So its four roots, counted with multiplicity, sum to 0. Three of them are 1, 2 and -4 , so the fourth is 1 again. A repeated root means tangency: the circle touches the parabola at $(1, 1)$ and meets it nowhere else. (Solving for D, E, F confirms the quartic is $(x-1)^2(x-2)(x+4)$.)

Question 8. Answer D.

Shift to $u = x - 2$ and $v = y - 3$, so that $x + y - 5 = u + v$ and the curve is $|u| + |v| + |u + v| = 4$. If u and v have the same sign, $|u + v| = |u| + |v|$ and the curve is $|u| + |v| = 2$. If they have opposite signs, $|u| + |v| + |u + v| = 2\max(|u|, |v|)$, and the curve is $\max(|u|, |v|) = 2$. So the region is a hexagon made of two triangles of area 2 (in the same-sign quadrants) and two 2×2 squares (in the opposite-sign quadrants): area $2 + 2 + 4 + 4 = 12$.

Question 9. Answer A.

The bisector divides AB in the ratio $CA : CB = 6 : 5$, so $BD = \frac{35}{11}$. Apply the cosine rule twice at B : $\cos B = \frac{25+49-36}{70} = \frac{19}{35}$, and then

$$CD^2 = 25 + \left(\frac{35}{11}\right)^2 - 2 \cdot 5 \cdot \frac{35}{11} \cdot \frac{19}{35} = 25 + \frac{1225}{121} - \frac{190}{11} = \frac{3025+1225-2090}{121} = \frac{2160}{121}.$$

So $CD = \frac{\sqrt{2160}}{11} = \frac{12\sqrt{15}}{11}$. The shortcut $CD^2 = ab \left(1 - \frac{c^2}{(a+b)^2}\right) = 30 \cdot \frac{72}{121}$ gives the same, and is worth deriving once.

Question 10. Answer C.

Rationalise the bracket: $\frac{1}{\sqrt{k}+\sqrt{k+1}} = \sqrt{k+1} - \sqrt{k}$. Then each term is

$$\frac{\sqrt{k+1} - \sqrt{k}}{\sqrt{k}\sqrt{k+1}} = \frac{1}{\sqrt{k}} - \frac{1}{\sqrt{k+1}},$$

and the sum telescopes to $1 - \frac{1}{\sqrt{100}} = \frac{9}{10}$.

Question 11. Answer E.

Each root satisfies $r^4 = 3r^2 - r - 1$, so $r^5 = 3r^3 - r^2 - r$. With $s_k = \sum r_i^k$: from Vieta, $s_1 = 0$ and the sum of products in pairs is -3 , so $s_2 = 0 - 2(-3) = 6$. The sum of products in threes is -1 , and $s_3 = s_1^3 - 3s_1e_2 + 3e_3 = 3(-1) = -3$. Hence

$$s_5 = 3s_3 - s_2 - s_1 = -9 - 6 - 0 = -15.$$

(Along the way $s_4 = 3s_2 - s_1 - 4 = 14$, option C.) No root is ever found.

Question 12. Answer B.

The tangent at $x = t$ is $y = 2tx - t^2$, which passes through $(1, -3)$ when $t^2 - 2t - 3 = 0$, so $t = -1$ or $t = 3$. The tangents are $y = -2x - 1$ and $y = 6x - 9$, meeting at $x = 1$. The parabola minus each tangent is a perfect square, $(x + 1)^2$ and $(x - 3)^2$ respectively, so the area is

$$\int_{-1}^1 (x + 1)^2 dx + \int_1^3 (x - 3)^2 dx = \frac{8}{3} + \frac{8}{3} = \frac{16}{3}.$$

In general the area for the point (p, q) is $\frac{2}{3}(p^2 - q)^{3/2}$; here $p^2 - q = 4$.

Question 13. Answer A.

Factorise the denominator as a difference of squares: $k^4 + k^2 + 1 = (k^2 + 1)^2 - k^2 = (k^2 - k + 1)(k^2 + k + 1)$. The two factors differ by $2k$, so

$$\frac{k}{k^4 + k^2 + 1} = \frac{1}{2} \left(\frac{1}{k^2 - k + 1} - \frac{1}{k^2 + k + 1} \right),$$

and $k^2 + k + 1$ is the next value of $k^2 - k + 1$. The sum telescopes to $\frac{1}{2} \left(1 - \frac{1}{111} \right) = \frac{55}{111}$.

Question 14. Answer C.

Read each root as the length of a segment: from $(0, 0)$ to $(x, 1)$, then from $(x, 1)$ to $(y, 3)$, then from $(y, 3)$ to $(6, 6)$. The three segments form a path from $(0, 0)$ to $(6, 6)$, so their total length is at least the straight-line distance $\sqrt{36 + 36} = 6\sqrt{2}$. Equality needs the points collinear on $y = x$, which happens at $x = 1$, $y = 3$. So the minimum is $6\sqrt{2}$. Calculus in two variables would get there too, far more slowly.

Question 15. Answer D.

We need $x > 0$, and then only the inequality involving the longest side matters. If $x > 1$, the longest side is x^3 , and we need $x + x^2 > x^3$, that is $x^2 - x - 1 < 0$, so $x < \frac{1+\sqrt{5}}{2}$. If $x < 1$, the longest side is x , and we need $x^2 + x^3 > x$, that is $x^2 + x - 1 > 0$, so $x > \frac{\sqrt{5}-1}{2}$. At $x = 1$ the triangle is equilateral. The answer interval is symmetric under $x \mapsto \frac{1}{x}$, as it must be: scaling the sides x, x^2, x^3 by x^{-4} gives x^{-3}, x^{-2}, x^{-1} .

Question 16. Answer F.

Let $S = \sum \frac{n^2}{2^n}$ and $T = \sum \frac{n}{2^n}$. Shifting the index, $\frac{S}{2} = \sum \frac{(n-1)^2}{2^n}$, so $S - \frac{S}{2} = \sum \frac{2n-1}{2^n} = 2T - 1$. Similarly $T - \frac{T}{2} = \sum \frac{1}{2^n} = 1$, so $T = 2$. Then $\frac{S}{2} = 3$ and $S = 6$. The subtraction trick removes one power of n at a time.

Question 17. Answer E.

Both curves are symmetric in the y -axis, so work with $x \geq 0$ and double. There, $4 - x^2 = x + 2$ at $x = 1$ and $x^2 - 4 = x + 2$ at $x = 3$. On $[0, 1]$ the curve is above the line; on $[1, 3]$ the line is above the curve, which dips to 0 at $x = 2$. The areas for $x \geq 0$ are $\int_0^1 (2 - x - x^2) dx = \frac{7}{6}$, $\int_1^2 (x^2 + x - 2) dx = \frac{11}{6}$ and $\int_2^3 (6 + x - x^2) dx = \frac{13}{6}$, totalling $\frac{31}{6}$. Doubling gives $\frac{31}{3}$. The first region straddles the y -axis as one piece, which does not affect the total.

Question 18. Answer B.

If P has x -coordinate a , then $x^3 - (3a^2x - 2a^3) = (x - a)^2(x + 2a)$, so Q is at $x = -2a$, and likewise R is at $x = 4a$. The region between the curve and the tangent at $x = t$ has area $\int (x - t)^2(x + 2t) dx$ over an interval of length $3|t|$, which scales as t^4 : it equals $\frac{27}{4}t^4$. So the areas are $\frac{27}{4}a^4$ and $\frac{27}{4}(2a)^4$, in the ratio $\frac{1}{16}$, whatever a is. Scaling does all the work once the first area is seen to be a fixed multiple of t^4 .

Question 19. Answer C.

Picture the line $y = x + k$, of gradient 1, sliding upwards across the graph of $y = |x^2 - 4x|$, which is zero at $x = 0$ and $x = 4$ and has a hump on $[0, 4]$ with peak 4 at $x = 2$. The outer arms $y = x^2 - 4x$ are steeper than 1 far out, so the line always crosses the right arm once when $k > -4$ and the left arm once when $k > 0$. The line meets the hump $y = 4x - x^2$ where $x^2 - 3x + k = 0$: twice when $0 < k < \frac{9}{4}$ (both roots in $[0, 4]$), once, tangentially, at $k = \frac{9}{4}$, and not at all beyond. So there are four solutions exactly for $0 < k < \frac{9}{4}$. At $k = 0$ the line passes through the corner $(0, 0)$ and there are three, and at $k = \frac{9}{4}$ there are three.

Question 20. Answer D.

The minimum of a sum of distances is attained at a median of $1, 2, \dots, n$. Pairing the outermost terms, $|x - 1| + |x - n| \geq n - 1$, $|x - 2| + |x - (n - 1)| \geq n - 3$, and so on, with equality at the median. So $m(n) = (n - 1) + (n - 3) + \dots$, which is $\frac{n^2}{4}$ for even n and $\frac{n^2 - 1}{4}$ for odd n . For even $n = 2m$, $m(n) = m^2$ is always a square: 25 values. For odd $n = 2m + 1$, $m(n) = m(m + 1)$ lies strictly between m^2 and $(m + 1)^2$ when $m \geq 1$, so it is a square only when $m = 0$, that is $n = 1$ with $m(1) = 0$. The total is 26.